Adsorption Thermodynamics — Session 1

Equilibrium, Langmuir/Henry, Temperature Dependence, Heat of Adsorption

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Learning goals (conceptual)

By the end of this session, you should be able to:

- Explain why adsorption equilibrium is governed by equality of chemical potentials μ with a reservoir at fixed T.
- Describe adsorption as a form of diffusive equilibrium between gas and adsorbed phase.
- Explain qualitatively how entropy (accessible volume) and energy (binding strength) both influence adsorption.

Learning goals (quantitative)

You should also be able to:

- Write and interpret the Langmuir isotherm and its Henry (low-pressure) limit.
- Connect the Henry coefficient K_{H} to excess chemical potential and an "insertion" picture.
- Explain how the heat of adsorption q_{ads} controls the temperature dependence of adsorption.
- Distinguish differential vs. isosteric heats of adsorption and understand their boundary conditions.

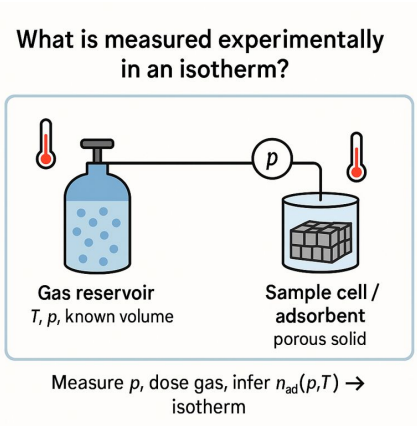
Roadmap for Session 1

1. Part A – Equilibrium & Chemical Potential
2. Part B – Langmuir Isotherm & Henry Coefficient
3. Part C – Henry Coefficient Intuition
4. Part D – Temperature Dependence
5. Part E – Heat of Adsorption
6. Wrap-up

Part A – Equilibrium & Chemical Potential

Experimental picture: adsorption isotherm

- A volumetric adsorption experiment:
  - Gas reservoir at known T and p,
  - Sample cell containing the porous solid,
  - Valves, pressure sensors, temperature control.
- We progressively dose gas and let the system equilibrate.



What is actually measured?

- For each step:
  - Initial and final gas pressures,
  - Known gas volume,
  - Temperature T.
- From the gas mass balance we infer:
  - Amount of gas remaining in the gas phase,
  - Amount taken up by the solid at that (T, p).
- Plotting uptake vs. pressure (at fixed T) gives an **adsorption isotherm**.

What are the right variables for adsorption?

- In a bulk gas, pressure p is natural: the system can expand or contract.
- In a *rigid* porous solid, volume is essentially fixed.
- A scalar "pressure inside the solid" is therefore not very meaningful.
- Instead, the natural variable for particle exchange is the **chemical potential** μ.

Diffusive equilibrium

Key idea

Adsorption is an example of **diffusive equilibrium**: molecules can move between gas and adsorbed phase until the chemical potentials match.

- At equilibrium:

$$\mu_{\text{gas}}(T, p) = \mu_{\text{ads}}(T, \sigma).$$

- Changing p at fixed T changes μ_{gas}, and therefore the equilibrium loading in the solid.

Thermodynamic pressure and rigidity (1)

- Thermodynamic pressure is defined by

$$p = -\left(\frac{\partial F}{\partial V}\right)_{T,N},$$

  where F is the Helmholtz free energy.

- This is a derivative with respect to volume at fixed T and N.

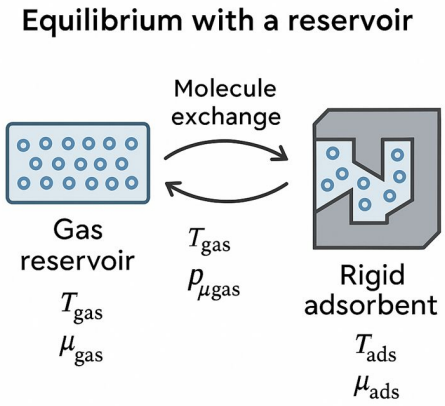
Thermodynamic pressure and rigidity (2)

- In a rigid solid:
  - The total volume changes only extremely slightly with loading,
  - Mechanical response is anisotropic (stress tensor).
- A scalar "internal pressure" is therefore not very useful.
- We instead treat the solid as a fixed volume and describe adsorption through μ and the number of adsorbed molecules.

Equilibrium with a reservoir

- Rigid adsorbent in contact with a gas reservoir.
- Reservoir fixes:
  - Temperature T_{gas},
  - Chemical potential $\mu_{\text{gas}}(T, p)$.
- At equilibrium:

$$T_{\text{ads}} = T_{\text{gas}}, \qquad \mu_{\text{ads}} = \mu_{\text{gas}}.$$



Take-home message of Part A

Adsorption as diffusive equilibrium

- The adsorbed phase and gas share the same T and μ at equilibrium.
- Changing p changes $\mu_{\text{gas}}(T, p)$ and thus the equilibrium loading.
- For rigid solids, we avoid "internal pressure" and track μ and N.

Bridge to simulation

Connection to GCMC (Session 2)

- Grand Canonical Monte Carlo works at fixed (T, μ, V) with fluctuating N.
- This is exactly the thermodynamic picture we just built: a rigid adsorbent exchanging molecules with a reservoir.

Part B – Langmuir Isotherm & Henry Coefficient

Langmuir model: assumptions

- A finite number of equivalent adsorption sites.
- Each site can be either empty or occupied by one molecule.
- No interaction between adsorbed molecules beyond site occupancy.
- Let θ be the fraction of occupied sites.

Langmuir adsorption isotherm

- Define:
  - σ: surface coverage,
  - $\sigma_{\max}$: saturation coverage,
  - b: equilibrium constant for adsorption.
- Langmuir isotherm:

$$\theta = \frac{\sigma}{\sigma_{\max}} = \frac{bp}{1 + bp}.$$

Interpretation

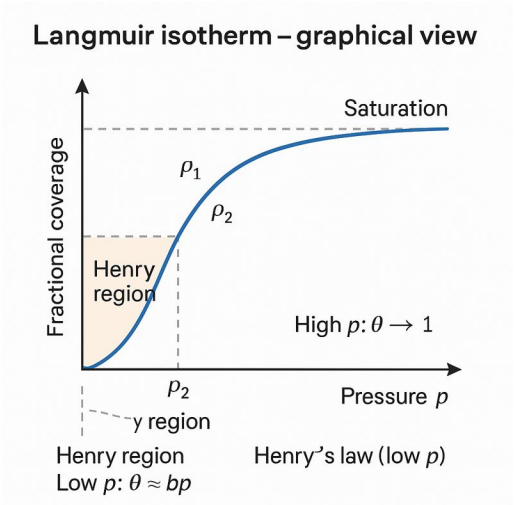
- Large b: strong adsorption, early rise, fast saturation.
- Small b: weak adsorption, gentle increase with p.

Langmuir isotherm – graphical view

- Low p:

$$\theta \approx bp \quad \Rightarrow \quad \sigma \approx \sigma_{\max} bp.$$

- This linear region corresponds to **Henry's law**.
- High p: sites saturate and $\theta \to 1$.



Chemical potentials: ideal gas

- For an ideal gas:

$$\mu^{\mathrm{IG}}(T, p) = \mu^{0}(T) + k_{\mathrm{B}} T \ln p.$$

- $\mu^{0}(T)$ is a reference chemical potential at some standard pressure.

Chemical potentials: adsorbed phase

- For the adsorbed phase:

$$\mu^{\mathrm{ads}}(T,\sigma) = \mu^{0}_{\mathrm{ads}}(T) + k_{\mathrm{B}}T\ln\sigma + \mu^{\mathrm{ex}}(T,\sigma).$$

- Terms:
  - $\mu^{0}_{\mathrm{ads}}(T)$: reference term,
  - $k_{\mathrm{B}}T\ln\sigma$: ideal mixing/entropy,
  - μ^{ex}: **excess** chemical potential (interactions + confinement).

From equilibrium to Henry coefficient (1)

$$\mu^{\mathrm{IG}}(T,p) = \mu^{0}(T) + k_{\mathrm{B}}\,T \ln p,$$
$$\mu^{\mathrm{ads}}(T,\sigma) = \mu^{0}_{\mathrm{ads}}(T) + k_{\mathrm{B}}\,T \ln \sigma + \mu^{\mathrm{ex}}(T,\sigma).$$

At equilibrium:

$$\mu^{\mathrm{IG}}(T,p) = \mu^{\mathrm{ads}}(T,\sigma).$$

From equilibrium to Henry coefficient (2)

Rearranging for coverage:

$$\sigma = \frac{p}{k_{\mathrm{B}} T} \exp\left[-\frac{\mu^{\mathrm{ex}}(T,\sigma)}{k_{\mathrm{B}} T}\right] \times \exp\left(-\frac{\Delta\mu^{0}(T)}{k_{\mathrm{B}} T}\right),$$

where $\Delta\mu^{0}(T) = \mu^{0}(T) - \mu^{0}_{\mathrm{ads}}(T)$ is absorbed into a prefactor.

- This is a general relation between σ and p that still contains the unknown $\mu^{\mathrm{ex}}(T,\sigma)$.

Henry regime and Henry coefficient

Low-loading limit

- As $\sigma \to 0$, interactions between adsorbed molecules vanish.
- The excess chemical potential becomes a constant:

$$\mu^{\mathrm{ex}}(T, \sigma) \to \mu^{\mathrm{ex}}(T, 0).$$

- Then the coverage becomes linear in pressure:

$$\sigma \approx K_{\mathrm{H}}(T)\, p.$$

Henry coefficient

$$K_{\mathrm{H}}(T) = \frac{1}{k_{\mathrm{B}} T} \exp\left[-\frac{\mu^{\mathrm{ex}}(T, 0)}{k_{\mathrm{B}} T}\right] \times (\text{constant factor}).$$

Part C – Henry Coefficient Intuition

Statistical mechanics view of K_{H} (1)

- In statistical mechanics,

$$K_{\mathrm{H}}(T) \propto \langle \exp[-\beta \Delta U(\mathbf{r})] \rangle_{\mathbf{r}} ,$$

where:

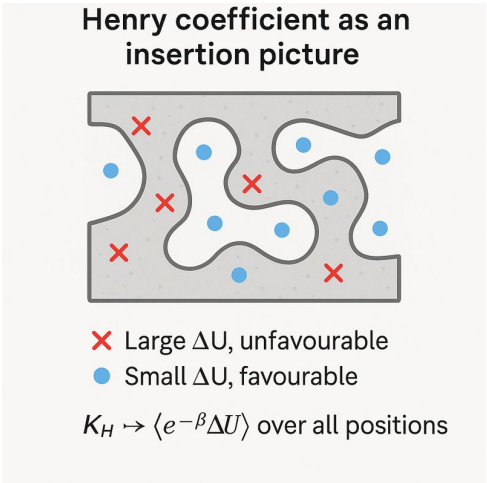
- $\beta = 1/(k_{\mathrm{B}} T)$,
- $\Delta U(\mathbf{r})$: energy change upon inserting a test molecule at position $\mathbf{r}$,
- The average is over random positions in the pore volume.

Statistical mechanics view of K_{H} (2)

- The insertion average combines:
  - **Accessible volume** (entropy) — how many positions give finite energy,
  - **Interaction energy** (enthalpy) — how deep the potential wells are.
- Both factors together determine whether K_{H} is large or small.

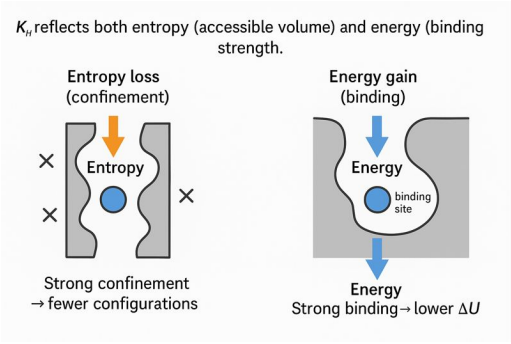
Henry coefficient as an insertion picture

- Imagine inserting a "ghost" molecule at many random positions in the pore.
- For each insertion, compute ΔU.
- Mostly repulsive positions:
  - ΔU large and positive,
  - $\exp(-\beta \Delta U)$ tiny,
  - K_{H} small.
- Many favourable positions:
  - ΔU modest or negative,
  - $\exp(-\beta \Delta U)$ sizeable,
  - K_{H} large.



Entropy vs energy in K_{H}

- **Entropic effect (confinement):**
  - Very narrow pores $\Rightarrow$ fewer accessible configurations,
  - Strong entropy loss $\Rightarrow$ lower K_{H}.
- **Energetic effect (binding):**
  - Strong attractive sites $\Rightarrow$ large energy gain,
  - Favourable positions boost K_{H}.



Design intuition from K_{H}

Structural trends

- Smaller pores $\Rightarrow$ stronger confinement and entropy loss $\Rightarrow$ often lower K_{H}.
- Strong binding sites (e.g. open metal sites) $\Rightarrow$ lower insertion energy $\Rightarrow$ higher K_{H}.
- At low T: energetics dominate.
- At high T: entropy and accessible volume dominate.

Part D – Temperature Dependence

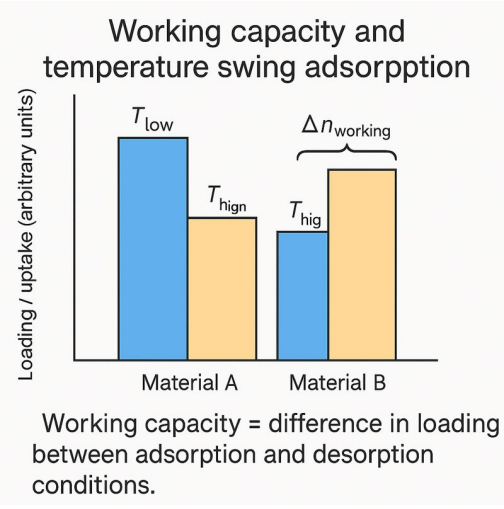
Working capacity and TSA

- In many processes we care about **working capacity**:

$$\Delta n = n(T_{\text{low}}) - n(T_{\text{high}}).$$

- Temperature swing adsorption (TSA):
  - Adsorb at T_{low} (high loading),
  - Desorb at T_{high} (low loading).
- The magnitude of Δn is governed by the heat of adsorption q_{ads}.

Working capacity – illustrative view



Temperature dependence of K_{H}

- At low loadings, a van't Hoff-like relation often holds:

$$\frac{\mathrm{d}\ln K}{\mathrm{d}T} = \frac{q_{\mathrm{ads}}}{k_{\mathrm{B}}T^2}.$$

- Here q_{ads} is the (differential) heat of adsorption at infinite dilution:

$$q_{\mathrm{ads}} \equiv h_{\mathrm{ads}} - h_{\mathrm{IG}}.$$

- h_{ads}: enthalpy per molecule in the adsorbed phase; h_{IG}: enthalpy per molecule in the gas phase.

Integrated temperature dependence

Starting from

$$\frac{\mathrm{d}\ln K}{\mathrm{d}T} = \frac{q_{\mathrm{ads}}}{k_{\mathrm{B}}T^2},$$

and assuming constant q_{ads}, integrate between T_0 and T:

$$\int_{K_0}^{K} \frac{\mathrm{d}K'}{K'} = \int_{T_0}^{T} \frac{q_{\mathrm{ads}}}{k_{\mathrm{B}}T'^2}\,\mathrm{d}T'.$$

This gives

$$\ln\left(\frac{K}{K_0}\right) = -\frac{q_{\mathrm{ads}}}{k_{\mathrm{B}}}\left(\frac{1}{T} - \frac{1}{T_0}\right),$$

or

$$K(T) = K_0 \exp\left[-\frac{q_{\mathrm{ads}}}{k_{\mathrm{B}}}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right].$$

Practical message: temperature dependence

Key implications

- Larger $|q_{\mathrm{ads}}| \Rightarrow$ stronger temperature dependence of K_{H}.
- Strongly binding materials give large working capacities in TSA, but may require more energy to regenerate.
- Weakly binding materials have small $|q_{\mathrm{ads}}|$ and weaker temperature sensitivity.

Part E – Heat of Adsorption

What do we mean by “heat of adsorption”? (1)

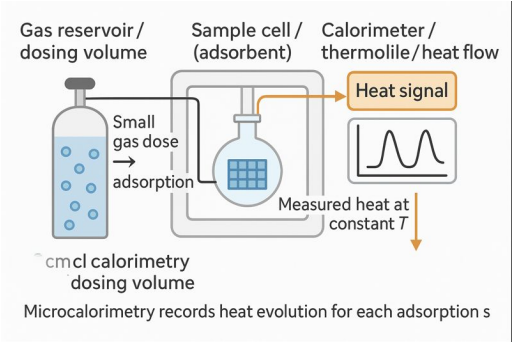
- When a molecule adsorbs, energy is typically released (exothermic).
- This heat must be removed to keep the sample at constant temperature.
- The measured heat per adsorbed molecule is called the **heat of adsorption**.

What do we mean by “heat of adsorption”? (2)

- Experimentally, this is often probed with **microcalorimetry**:
  - Monitor heat flow while dosing small amounts of gas.
  - Keep the system isothermal.
- Conceptually, we can define heats under different boundary conditions:
  - Constant T, V: differential heat,
  - Constant T, p: isosteric heat.

Schematic microcalorimetry experiment

- Gas reservoir delivers small doses into a cell with adsorbent.
- Each dose causes some adsorption.
- Calorimeter records the heat evolved to keep T constant.
- From heat vs. uptake we obtain q_{ads}(loading).



Differential heat (constant T, V)

- Defined at fixed temperature and volume:

$$q_{\mathrm{diff}}(T, V) \approx -\left(\frac{\partial U}{\partial N}\right)_{T,V}.$$

- Focuses on changes in **internal energy** of adsorbent + adsorbate.
- Natural quantity in simulations performed at constant T, V.

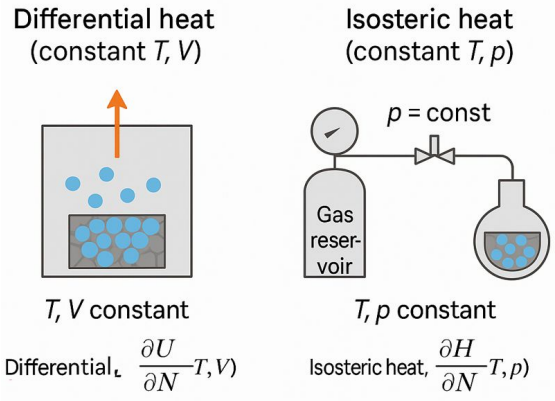
Isosteric heat (constant T, p)

- Defined at fixed temperature and external pressure:

$$q_{\mathrm{st}}(T, p) \approx -\left(\frac{\partial H}{\partial N}\right)_{T,p}.$$

- Related to changes in **enthalpy**.
- Often extracted from isotherms measured at different temperatures using Clausius–Clapeyron-like relations.

Differential vs isosteric heats – schematic



Same material, different boundary conditions → potentially different numerical 'heat of adsorption'

Why boundary conditions matter

Important cautions

- The same material can give different numerical "heats of adsorption" depending on what is held fixed in experiment or simulation.
- Always check:
  - Is q_{ads} defined at constant T, V or constant T, p?
  - Is it measured at low loading (Henry regime) or at higher coverage?
- To compare simulation with experiment, match the boundary conditions.

Wrap-up

Session 1 – Key takeaways (1)

1. In rigid porous solids, "pressure inside the solid" is not well-defined, so we describe adsorption using the chemical potential μ.
2. Adsorption is a **diffusive equilibrium**: at equilibrium, T and μ are equal in gas and adsorbed phases.
3. The Langmuir isotherm provides a simple picture with saturation behaviour.

Session 1 – Key takeaways (2)

4. The Henry coefficient K_{H} is controlled by the excess chemical potential at infinite dilution: $K_{\mathrm{H}} \propto \exp(-\mu^{\mathrm{ex}}/k_{\mathrm{B}}T)$.
5. Temperature dependence of K_{H} follows a van't Hoff-like relation governed by the heat of adsorption q_{ads}.
6. "Heat of adsorption" depends on boundary conditions (differential T, V vs isosteric T, p); always check which one is used.

Looking ahead: Session 2

- We now have the minimal thermodynamics needed for adsorption:
  - Role of μ,
  - Langmuir and Henry behaviour,
  - Temperature dependence and heats of adsorption.
- Next session:
  - Statistical thermodynamics of adsorption,
  - Grand Canonical Monte Carlo (μVT) ensemble,
  - Connecting simulation outputs to these thermodynamic ideas.